

Shaojie Chen

305-310-5596 | cheerioov2@gmail.com

SUMMARY

Application / Systems Engineer supporting CHP, distributed generation, and microgrid projects with hands-on experience in reciprocating engine-generators, hybrid inverter-BESS/HEMS integration, BOP/protection, ComAp controls, Modbus/RS-485, DAQ validation, and technical documentation. Translate site and system requirements into electrical/thermal architectures, vendor requirements, commissioning workflows, and customer-ready technical-commercial proposals while coordinating international manufacturers, sales teams, customers, and management.

WORK EXPERIENCE

Enginuity Power Systems

Detroit, MI

Senior Project Engineer

June 2024 - Present

- Support CHP and microgrid architecture definition for residential and commercial DER concepts, including reciprocating engine-generator integration, hybrid inverter/BESS interfaces, controller selection, breaker/protection concepts, CT metering, Modbus/RS-485 communication, enclosure ventilation, gas sensing, and serviceability requirements.
- Lead system design and controls integration for a residential HEMS coordinating engine generation, hybrid inverter operation, battery storage, smart loads, domestic hot water demand, space-heating profiles, and real-time electrical consumption.
- Develop electrical and thermal load-emulation capability to reproduce residential power, domestic-hot-water, and space-heating demand profiles for repeatable micro-CHP validation, controls calibration, transient testing, and troubleshooting.
- Evaluate hybrid inverter and battery behavior using high-speed DAQ and power analysis, characterizing waveform quality, transfer timing, bypass capability, efficiency trends, current ripple, and grid-parallel / backup operating behavior.
- Coordinate with international generator, inverter, battery, controller, and electrical-equipment manufacturers and suppliers to align specifications, evaluate integration constraints, compare BOP vendors, resolve issues, and reduce deployment risk before release.
- Develop commissioning and remote-connectivity documentation for ComAp IG500/IG1000-based CHP systems, covering 4G/LAN connectivity, WebSupervisor enrollment, firmware/archive updates, remote diagnostics, and field handover requirements.
- Support early-stage project development and customer-facing proposals by translating test data, supplier quotations, site constraints, and system requirements into configurations, budget assumptions, scope clarifications, and RFP comments/exceptions for sales and management review.
- Resolved engine ignition reliability issues on an OP4S reciprocating engine through custom long-reach spark plug development, iterative testing, supplier coordination, and material validation.

Development Engineer

April 2023 - June 2024

- Executed R&D test plans for the 8 kW micro-CHP platform, built data-driven validation routines, and accelerated project timelines.
- Upgraded test equipment with programmable electrical and thermal load banks, including system design, PCB development, prototyping, and validation to expand automated test capability.
- Enhanced embedded control systems using CLICK PLC and Arduino/ESP32 platforms, implementing safety interlocks, limit logic, auxiliary device control, and cleaner code structures for safer operation.
- Coordinated testing and technical specification alignment with international generator and inverter manufacturers to support system integration, efficiency evaluation, and validation of electrical interfaces.

Patent: METHODS AND DEVICES FOR SELECTING A POWER SOURCE (509846268)

RELEVANT CHP / DER APPLICATION ENGINEERING PROJECTS

- **Microgrid Controls Architecture:** Evaluated ComAp IntelliGen and IntelliLite controller strategies for CHP and microgrid use cases, including synchronization, grid-parallel/island operation, breaker control, remote commissioning, and third-party Modbus integration with hybrid inverters and BMS.
- **BOP / System Definition:** Supported 25-50 kW and 200 kW CHP/microgrid concept development, including controller I/O allocation, protection philosophy, CT selection, contactor and motorized breaker selection, enclosure ventilation, gas sensing, serviceability, and supplier interface requirements.
- **Power Quality / Inverter Validation:** Characterized ATS transfer time, bypass limits, current ripple, inverter efficiency, backup behavior, and DC/AC interface constraints using Dewesoft DAQ, differential voltage probes, and current clamps.
- **Field Connectivity / Commissioning:** Authored field-deployable procedures for ComAp IG500/IG1000 connectivity, WebSupervisor setup, remote diagnostics, firmware/archive update workflows, and site handover support.

EDUCATION

University of Miami

Miami, FL

Master of Science in Mechanical Engineering

Dec 2022

Relevant Courses: Additive Manufacturing, Finite Element Method, Construction Management, Management of Technology, Ergonomics and Human Factors Engineering

Bachelor of Science in Mechanical Engineering

May 2021

Relevant Courses: Internal Combustion Engine, Fluid Analysis, System Dynamics, Heat Transfer, Mechanics of Solids, Thermodynamics

SKILLS

Technical-Commercial: RFP review, budgetary proposals, BOP configuration, supplier quotations/vendor comparison, project economics, customer meetings/presentations, international manufacturer interface, negotiation, and cross-functional communication at customer and management levels
Power / Controls: CHP/DG/DER integration, reciprocating engine-generator systems, heat recovery, hybrid inverter-BESS integration, HEMS, ComAp genset controllers, CLICK PLC, CAN bus, Modbus/RS-485, Dewesoft DAQ/power analysis, electrical and thermal load banks
Software / Tools: SolidWorks, Creo, AutoCAD, STAR-CCM+, MATLAB, Python, Arduino/ESP32, PLC programming, technical documentation, root-cause analysis, DFM/DFA; Languages: Chinese (Native), English (Advanced)